

Technical Datasheet — T4R1FLASH

Case: `t4\_r1\_flash` | Generation date: 2026-08-25 | Doc: VBF-T4R1FLASH-DSH-01

All values mechanical from simulation outputs / parameter set (see log.jsonl); no numbers from memory.

Field	Value	Source
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Rated capacity	3.93 Ah nominal; 3.955 Ah verified at 1C, 25 degC	parameter set + cell/final_b7_1c_dfn.json
Nominal voltage / window	3.814 V midpoint; 2.5–4.2 V	cell/final_b7_energy_dfn.json + parameter set
Rated energy	14.04 Wh (integral of V·I over 1C discharge)	cell/final_b7_energy_dfn.json
Gravimetric energy density	460.1 Wh/kg (contract caliber, electrolyte excluded)	cell/final_b7_energy_dfn.json
Volumetric energy density	899.3 Wh/L (contract caliber)	cell/final_b7_energy_dfn.json
Maximum continuous discharge	1C verified (3.955 Ah delivered); higher rates not simulated	cell/final_b7_1c_dfn.json
Fast-charge capability	4C at 45 degC: T_max 329.6 K (56.5 degC), no lithium plating (anode min +0.0193 V)	cell/final_b7_4c45_dfn.json
Operating temperature range	-20 degC (retention 0.9929 at 1C) to 45 degC (4C exam ambient), per simulated conditions	cell/final_b7_lowt_dfn.json, cell/final_b7_4c45_dfn.json
Cycle life	**Not simulated (requires aging model)**	—
Thermal runaway termination	No plating; T_max within red line; overcharge (0.5C to 4.7 V) does not trigger thermal runaway (triggered=False)	cell/final_b7_4c45_dfn.json, cell/final_b7_lowt_dfn.json
Dimensions / mass	Electrode strip 65 x 1580 mm; stack thickness 152 µm; mass 30.5 g (contract caliber, excl. electrolyte) / ~35.5 g incl. electrolyte	parameter set + calc-e
DC resistance	3.85 mOhm (1C, 50% SOC estimate)	cell/final_b7_energy_dfn.json